

Dataset Inspection Report – Telco Customer Churn Dataset

1. Dataset Overview

The Telco Customer Churn dataset contains information about customers of a telecommunications company. It includes customer demographics, services subscribed to, account information, billing details, and churn status. The dataset is used to analyze customer behavior and identify factors contributing to customer churn.

2. Dataset Summary

Attribute	Value
Dataset Name	Telco Customer Churn
Source	IBM sample Dataset (via Kaggle)
Number of Rows	7,043
Number of columns	21
Target Variable	Churn
File Format	CSV

3. Data Types

The dataset contains a mix of:

- . Categorical variables: Gender, Partner, Dependents, Phone Service, Internet Service, Contract, Payment Method, Churn.
- . Numerical variables: Tenure, Monthly Charges, Total charges.

4. Data Quality Assessment

The dataset was inspected before analysis.

Check	Result
Missing Values	None
Duplicate Records	None
Blank Cells	None
Incorrect Data Types	None
Data ready for Analysis	Yes

5. Key Variables Used

Variable	Description
Customer ID	Unique customer identifier
Gender	Customer gender
Senior Citizen	Indicates whether the customer is a senior citizen
Partner	Whether the customer has a partner
Dependents	Whether the customer has dependents
Tenure	Number of months the customer has stayed with the company
Internet Service	Type of internet service
Contract	Contract type
Payment Method	Customer payment method
Monthly Charges	Monthly bill amount
Total Charges	Total amount paid by customer
Churn	Indicates whether the customer left the company

6. Initial Observations

The inspection confirmed that the dataset was clean and suitable for analysis. No missing values or duplicate records were found. The data included a combination of categorical and numerical variables, making it appropriate for exploratory analysis and dashboard creation in Excel and Power BI.

7. Conclusion

The dataset met the quality requirements for analysis. Its completeness and structure allowed for reliable exploration of customer behavior and churn patterns. The inspection stage ensured that subsequent analyses and business recommendations were based on accurate and consistent data.